Tyler Stowell
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EDUCATION

Brigham Young University – Bachelor of Science, Electrical Engineering Expected April 2027

- Minor: Physics
- Cumulative GPA: 3.99 / 4.0
- Classes in VLSI, Integrated Photonics, Quantum Optics, Cleanroom Fabrication, Semiconductor Devices, Optoelectronics, RF Design, and Signal Processing

EXPERIENCE

CamachoLab - Research Assistant Sep 2021 - Jun 2022, Jun 2024 - Present

- Designed, fabricated, packaged, and characterized integrated silicon photonics devices in collaboration with Nokia engineers
- Deployed a quantum random number generator on site at Oak Ridge National Laboratory for a quantum key distribution system
- Wire bonded and edge coupled photonic chips in BYU's cleanroom
- Designed couplers, crossover waveguides, and modulator setups for an active photonic integrated circuit (PIC) with collaborators at the University of Arizona, University of Maryland, and Tyndall National Institute

Center for Quantum Networks - Quantum Engineer (Contract) May 2026 - Aug 2026

- Designed, built, and characterized a quantum frequency conversion system for SiV quantum memory experiments, achieving ~75% internal quantum efficiency
- Developed a cavity-coupled color center simulator that reproduces published experimental results to prepare for experimental work on a system ordered from IonQ (manuscript in preparation)
- Restored a pulsed laser system, including water-cooling integration, nitrogen purging, and optical alignment; integrated for support of the SiV quantum memory experiment.

Harsch Systems - Design Engineer June 2020 - Aug 2021

- Designed two generations of an RP2040-based PCB for industrial control of 4-20 mA and 0-10 V inputs and outputs
- Wrote a Linux kernel driver so Raspberry Pi based controllers could meet strict SPI timing requirements for industrial I/O applications under heavy load
- Brought up embedded Rust on the then-unsupported RP2040, writing register level GPIO, ADC, and PWM test programs

PUBLICATIONS

Integrated Differential Conjugate Homodyne Detection for Quantum Random Number Generation – arXiv Preprint (Under review)

- Wrote FPGA code for Toeplitz hash-based entropy extraction from quantum shot-noise limited measurements at a rate of 12 Mbps

Quantum Conference Key Agreement Using a Photonic Integrated Circuit Green Machine – Conference Article (Optica Quantum 2.0)

- Characterized a PIC capable of superadditive measurements of quantum signals
- Built a digital twin and QKD simulations (decoy states, finite-key effects) to compute conference key rates exceeding GHZ based protocols

Device-Aware Quantum Photonic Simulator for Gaussian States – Conference Paper (Frontiers in Optics + Laser Science 2022)

- Contributed quantum simulation features to the Simphony photonic circuit simulator, eventually extending it to sums of Gaussian states (e.g. cat states)

AWARDS & HONORS

1st Place – Quantum Rings IQC Hackathon (2026)

- Led a team to a first-place win in the Quantum Rings challenge for the first Intermountain Quantum Competition
- Developed a way to predict the runtime of quantum circuits with up to 250 qubits and 6 million gates using a tensor network probe method
- Outperformed a previous industry model on their own metrics (94% compared to 82%)

SELECTED PROJECTS

Single-Wheel Robot (2025)

- Designed, 3D-printed, wired, and programmed a modular robot from scratch that converts between two-wheel and single-wheel balanced modes (hobby, not class project)

Laser Tag System (2026)

- Designed an optical transceiver board (LED transmitter, photodiode receiver, filtered amplifier) that placed in the top 10% of the class for signal-to-noise ratio

Real-Time Noise Floor Computer (2021)

- Wrote code that computed the noise floor of signals from the adaptive-optics wavefront sensor on the Daniel K. Inouye Solar Telescope in real time

SKILLS

Photonics and Optical Engineering

- Lumerical FDTD, GDSFactory, silicon photonics, free-space optics, optical alignment, fiber coupling, PIC characterization and packaging

Electrical and Hardware Engineering

- Cadence, KiCAD, FPGA development (Xilinx Vivado), PCB Design, wire bonding, cleanroom fabrication, CNC machining

Programming Languages

- Python, C, MATLAB, Rust, Linux kernel development, quantum simulation (qutip, qiskit)

Languages

- English (native), Spanish (fluent)

SERVICE & LEADERSHIP

The Church of Jesus Christ of Latter-day Saints – Volunteer July 2022 – June 2024

- Served a two-year, Spanish-speaking volunteer mission in southern Argentina, teaching and doing community service
- Led and trained groups of volunteers as district and zone leader
- Served as the primary financial and housing secretary for 6 months, managing all finances and housing contracts for ~150 volunteers